LESSON PLAN

A/C ENGINEERING SYSTEM (17)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	26
Total No. of Lesson Plans	30	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Types of oxygen systems, oxygen storage, regulators and masks
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Types of oxygen systems, oxygen storage, regulators and masks
5. Specific Objectives	17.2.1.1 Trainee will be able to understand the need of oxygen systems, types of systems, working principles of components.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Identification of oxygen storage types, regulator components, and mask assemblies using L & Ds and IB; Safety precautions regarding oxygen purity and pressure limits; Instructor supervision of component identification.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) Why is an oxygen system essential for aircrew at high altitudes?
(b) What are the primary hazards associated with oxygen systems?
(c) Can you list three common methods used to store oxygen in aircraft?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Need for oxygen systems in aviation • Types of oxygen systems: High pressure vs Low pressure • Working principle of high pressure systems • Working principle of low pressure systems • Comparison of system types based on aircraft category	L & Ds	PPT, IB	15 min
(b)	• Oxygen storage methods: Cylinder vs Liquid • Regulator components and pressure reduction • Mask types: Demand vs Continuous flow • Function of pressure reducing valves • Safety features in oxygen delivery systems	L & Ds	PPT, PDF	20 min
(c)	• Identification of regulator parts • Operation of demand oxygen masks • Maintenance checks for oxygen systems • Emergency procedures for system failure • Troubleshooting common regulator faults	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between high-pressure and low-pressure oxygen systems.
(b) Describe the function of a pressure reducing valve in an oxygen system.
(c) What are the key safety precautions when handling oxygen cylinders?

12. Home Assignment and Follow Up	Prepare a brief note on the charging procedure for liquid oxygen systems to be discussed in the next lesson plan (LP 27).
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Liquid O2 system, O2 charging procedure	Refer to the master syllabus reference list	LP 27

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min